LAB 7: Fuzzing for Input Validation Bugs

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Original Application Output

Execution of the original processor.py with sample inputs, before fixing the validation bugs

```
PS C:\Users\umasa\Documents\L6_L7_02_car_rental_rate\Lab 7> python processor.py
==== Running sanitize_license_plate ====
Enter license plate with delimiters (-, _, #): KA-03-MH-1234
Sanitized Plate: KA03MH1234

==== Running parse_trip_mileage ====
Enter CSV of mileages (e.g. 100.5, 45.2, 80.0): 20.2
Parsed Mileage List: [20.2]

==== Running format_rental_invoice ====
Enter rental record (Driver:Days:Category): divyaa:9090:economy
Rental Invoice: Driver divyaa booked economy for 9090 days: Total $272700.00
PS C:\Users\umasa\Documents\L6_L7_02_car_rental_rate\Lab 7>
```

Hypothesis Fuzzing — Initial Failures

Property-based tests expose crashes caused by malformed and missing inputs

```
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
=========================================== short test summary info ============================================
FAILED test_processor.py::test_sanitize_license_plate_no_crash - AssertionError: sanitize_license_plate crashed with input
 None: 'NoneType' object has no attribute 'strip'
FAILED test_processor.py::test_parse_trip_mileage_safe - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exce
ptions)
FAILED test_processor.py::test_format_rental_invoice_safe - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-e
xceptions)
========================================== 3 failed, 1 warning in 17.48s ===========================================
PS C:\Users\umasa\Documents\L6_L7_02_car_rental_rate\Lab 7>
```

Fixed Application Output

Execution of the corrected processor.py with valid sample inputs

```
PS C:\Users\umasa\Documents\L6_L7_02_car_rental_rate\Lab 7> python processor.py
 ==== Running sanitize_license_plate ====
 Enter license plate with delimiters (-, _, #): KA-03-MH-1234
 Sanitized Plate: KA03MH1234

 ==== Running parse_trip_mileage ====
 Enter CSV of mileages (e.g. 100.5, 45.2, 80.0): 100.5, 45.2, 80.0
 Parsed Mileage List: [100.5, 45.2, 80.0]

 ==== Running format_rental_invoice ====
 Enter rental record (Driver:Days:Category): divyaa:5:economy
 Rental Invoice: Driver divyaa booked economy for 5 days: Total $150.00
PS C:\Users\umasa\Documents\L6_L7_02_car_rental_rate\Lab 7>
```

Hypothesis Fuzzing — Tests Passed

All three Hypothesis tests pass after the input-validation fixes

```
PS C:\Users\umasa\Documents\L6_L7_02_car_rental_rate\Lab 7> python -m pytest -v test_processor.py
rootdir: C:\Users\umasa\Documents\L6_L7_02_car_rental_rate\Lab 7
plugins: anyio-4.14.1, hypothesis-6.170.15, langsmith-0.14.16, asyncio-1.4.1, typeguard-4.6.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 3 items

test_processor.py::test_sanitize_license_plate_no_crash PASSED                                                    [ 33%]
test_processor.py::test_parse_trip_mileage_safe PASSED                                                            [ 66%]
test_processor.py::test_format_rental_invoice_safe PASSED                                                         [100%]

============================================== warnings summary ==============================================
..\..\..\AppData\Local\Programs\Python\Python312\Lib\site-packages\anyio\pytest_plugin.py:14
  C:\Users\umasa\AppData\Local\Programs\Python\Python312\Lib\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10War
ning: _pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
========================================== 3 passed, 1 warning in 3.18s ==========================================
PS C:\Users\umasa\Documents\L6_L7_02_car_rental_rate\Lab 7>
```